

EE624: Speech Technology Course Projects

Submission date: On or before April 30, 2026; Strict deadline with delay penalty: May 10, 2026

Project 1: Perform DTW based isolated vowel recognition

- Record **twenty five** utterances of each of the isolated English digits (0-9), vowels (a,e,i,o,u), and the given sentence (*I'm registered with speech technology course and recording this data for the course project*) in your voice. Record speech data over PC using Audacity software in PCM format with a sampling frequency of 16 kHz and a precision of 16-bits. Store the recorded data as: <Roll.No>_<digit>/<vowel>/sentence_<recording_num>.wav. Human voice characteristics usually differ slightly over the time and it is referred to as the session variation. So you are advised to record different entities at least in two sessions.
- For each of the vowels, choose one of the utterance as the reference template and the remaining ones as the test templates. Compute 39-dimensional MFCC features comprising base, delta, and delta-delta components for all the recordings using MATLAB. Using DTW command of MATLAB, write a script that computes DTW scores for any two similar or dissimilar test utterances and also plots the corresponding DTW curve.

Project 2: Perform GMM based isolated digit recognition

- Given the training and testing file lists, compute 39-dimensional MFCC features and learn a GMM having 16/32 mixtures for each of the isolated digits using training data.
- Identify the digits in test data using the maximum likelihood rule over the trained GMMs and report the confusion matrix for the recognition task.

Project 3: Develop context dependent subword GMM-HMM and DNN-HMM based ASR systems for given speech data using Kaldi toolkit

- (A). GMM-HMM based ASR systems for continuous speech recognition.
- Compute MFCC features for the given continuous speech database
 - Train mono-phone model for the vocabulary
 - Create tri-phone HMM models and train tied system with 8 Gaussian per state
 - Train a bigram LM using the training transcript
 - Perform Viterbi based decoding of the test set
 - Evaluate performance
- (B). Develop a hybrid DNN-HMM system for continuous speech recognition.
- Train a feed-forward neural network to map MFCC features to the triphone posterior probabilities produced by the GMM-HMM system developed in part (A)
 - Perform Viterbi based decoding of the test set using bigram LM
 - Evaluate performance

***** Deliverables for each course project will be pointed out during the course *****